Paper 60: Causal Purity as an Independent Control Parameter in an Ising Spin System

Cross-Substrate Verification of the Causal-Purity Universality Class

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Abstract

Papers 57–59 established that causal purity P_{causal} controls a new type of non-equilibrium order in VCML — distinct from temperature-driven phase transitions. The universality class claim requires the same mechanism to appear in a different physical substrate.

We test this in a 2D Ising spin system augmented with a VCSM-lite memory layer (slow field ϕ, updated by zone-attributed wave events, gated by causal purity). The Ising model has one standard order parameter (magnetisation m, controlled by temperature T); the VCSM layer adds a second order parameter ($S_\phi = |\bar{\phi}_{\text{zone0}} - \bar{\phi}_{\text{zone1}}|/\sigma_\phi$, controlled by P_{causal}).

Key result: m depends only on T (columns of the (T, P_{causal}) phase diagram are constant). S_ϕ depends monotonically on P_{causal} at *all* temperatures, including $T > T_c$ where spins are completely disordered ($m = 0.07$, S_ϕ ranges from 0.51 to 0.91).

The two order parameters are **orthogonal control parameters**: temperature controls spin order; causal purity controls memory order. The memory layer develops zone-specific structure even when the underlying spin substrate is thermally disordered — exactly the condition under which attractor persistence (ferromagnet-like) fails and reconstructive persistence (VCSM-like) is the only available mechanism.

This is the first cross-substrate demonstration that causal purity is an independent control parameter for non-equilibrium order. It supports the universality class hypothesis: VCSM describes a new class of purity-controlled stochastic field dynamics, not specific to VCML.

1 Introduction and Motivation

Paper 59 derived the Model A field equation for VCML's fieldM:

$$\partial_t \phi = -\kappa\phi + D\nabla^2\phi + r_w F A\, s(x) + \eta, \quad \langle \eta^2 \rangle \propto (1 - P_{\text{sp}})^2 V_{\text{bnd}} + (1 - P_{\text{temp}})^2 V_{\text{temp}}. \tag{1}$$

The noise amplitude is controlled by *causal purity*, not temperature. Most field theories use temperature as the noise control parameter; in VCML, causal purity plays that role.

For this to be a genuine universality class, the same purity-controlled phase transition must appear in different physical substrates. The simplest clean test: a 2D Ising ferromagnet (the prototype attractor persistence system) with a VCSM-lite memory layer on top.

If S_ϕ transitions with P_{causal} while m transitions with T — and the two are orthogonal — then causal purity is a substrate-independent control parameter for a new kind of non-equilibrium order.

2 The Ising + VCSM-lite Model

2.1 Ising substrate

A 40×40 lattice ($N = 1600$ sites) with periodic boundary conditions. Spins $s_i = \pm 1$, coupling $J = 1$, temperature T. Standard Metropolis dynamics (checkerboard sublattice sweeps). Two spatial zones: left half ($x < L/2$) = zone 0, right half = zone 1.

2.2 VCSM-lite memory layer

Each site i carries a slow memory variable ϕ_i updated by periodic zone-attributed wave events.

Wave mechanism. Every 25 steps, a wave launches from zone z (alternating $z = 0, 1$). The wave applies a zone-specific external field to sites within Manhattan radius $r = 5$:

$$h_{\text{ext}} = \begin{cases} +1.5 & \text{zone 0 wave} \quad (\text{biases spins toward } +1) \\ -1.5 & \text{zone 1 wave} \quad (\text{biases spins toward } -1) \end{cases} \tag{2}$$

for 5 Metropolis sweeps. The resulting spin deviation $\delta_i = s_i - b_i$ (where b_i is the exponentially-weighted spin baseline at rate $\beta = 0.005$) carries a zone-specific signal: $\delta_i > 0$ for zone-0 sites during a zone-0 wave, $\delta_i < 0$ for zone-1 sites during a zone-1 wave.

Causal gate. For each site in the wave radius, the contribution to the fast memory m_i is:

$$m_i \mathrel{+}= \text{FA} \times \begin{cases} \delta_i \cdot \text{sign}(z = z_i) & \text{with prob } P_{\text{causal}} \quad (\text{correct attribution}) \\ \mathcal{N}(0, \sigma_\delta) & \text{with prob } 1 - P_{\text{causal}} \quad (\text{noise}) \end{cases} \tag{3}$$

where z_i is the zone of site i. At $P_{\text{causal}} = 1$: zone-consistent signal always written. At $P_{\text{causal}} = 0$: pure noise.

Write gate. When a site's streak of same-sign spins reaches $SS = 8$, a write fires: $\phi_i \mathrel{+}= \text{FA}(m_i - \phi_i)$, followed by $\phi_i \mathrel{\times}= \text{FIELD_DECAY} = 0.999$ each step.

Order parameters.

$$m = \left|\langle s_i \rangle\right| \qquad \text{Ising magnetisation (spin order)} \tag{4}$$

$$S_\phi = \frac{|\bar{\phi}_{\text{zone0}} - \bar{\phi}_{\text{zone1}}|}{\sigma_\phi + \epsilon} \qquad \text{phi-order (memory order)} \tag{5}$$

3 Results

Table 1: Steady-state S_ϕ (phi-order). Rows = temperature; columns = P_{causal}. S_ϕ increases monotonically with P_{causal} at every temperature, including $T > T_c$ where $m \approx 0$.

T	0.50	0.60	0.70	0.75	0.80	0.85	0.90	0.95	1.00
1.00	0.12	0.15	0.17	0.18	0.20	0.21	0.23	0.26	0.28
1.50	0.35	0.40	0.45	0.46	0.49	0.51	0.53	0.55	0.57
2.00	0.37	0.42	0.47	0.49	0.52	0.54	0.56	0.58	0.60
2.27	0.43	0.50	0.58	0.60	0.64	0.67	0.70	0.74	0.77
3.00	**0.51**	**0.60**	**0.68**	**0.71**	**0.75**	**0.79**	**0.83**	**0.87**	**0.91**
4.00	0.32	0.38	0.45	0.48	0.51	0.53	0.58	0.61	0.64

Table 2: Magnetisation m (spin order). Columns are constant: m is T-controlled and P_{causal}-independent. Standard Ising transition visible at $T_c \approx 2.27$.

T	0.50	0.60	0.70	0.75	0.80	0.85	0.90	0.95	1.00
1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1.50	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
2.00	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
2.27	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54
3.00	**0.07**	**0.07**	**0.07**	**0.07**	**0.07**	**0.07**	**0.07**	**0.07**	**0.07**
4.00	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04

3.1 Finding 1: orthogonal control parameters

The magnetisation m varies across rows (temperatures) but not across columns (P_{causal} values): m is T-controlled and P_{causal}-independent. The phi-order S_ϕ varies across both rows and columns: S_ϕ depends on both T and P_{causal}, but the P_{causal} dependence is present at *every* temperature row (Figure **??**b,c).

The two order parameters have distinct control parameters. This is the definition of orthogonality: changing T does not change P_{causal}'s effect on S_ϕ, and vice versa.

3.2 Finding 2: phi-order exists at $\mathbf{T > T_c}$ (reconstructive beats attractor)

At $T = 3.0$ ($m = 0.07$, Ising completely disordered), S_ϕ ranges from 0.51 to 0.91 depending on P_{causal}. The memory layer maintains zone-specific order in ϕ even when the spin substrate is thermally disordered and supports no attractor persistence. This is direct evidence of the two-layer coexistence: the attractor layer (Ising) fails at $T > T_c$, but the reconstructive layer (VCSM-lite) succeeds as long as P_{causal} is sufficiently high.

3.3 Finding 3: $\mathbf{S_\phi}$ peaks near and above $\mathbf{T_c}$, not at low T

Counter-intuitively, S_ϕ is *lowest* at $T = 1.0$ (highly ordered spins, $m = 1.00$) and highest near and above T_c. At $T = 1.0$, spins are near their ground state: the wave perturbation cannot move them significantly, producing small deviations $\delta_i \approx 0$ and weak signal in m_i. At $T > T_c$, spins are susceptible to the wave's external field, producing large deviations and strong zone-specific signal in ϕ.

This mirrors the VCML finding that crystallisation (zero collapse rate) hurts persistence: the VCSM layer needs the carrier to be *perturbed* to write. An overly stable attractor (low-T Ising, crystallised VCML) starves the reconstructive layer of signal. Optimal operation requires non-zero substrate volatility.

4 Interpretation: Two Independent Phase Structures

The experiment reveals two independent phase structures in the augmented Ising system:

Structure 1 (standard physics): Ising ferromagnetic phase transition at $T_c \approx 2.27$. Below T_c: $m > 0$ (spin order, attractor persistence). Above T_c: $m \approx 0$ (spin disorder, no attractor persistence). Control parameter: T.

Structure 2 (new): Purity-controlled phi-order. S_ϕ increases monotonically with P_{causal} at all temperatures. $S_\phi > 0$ requires only $P_{\text{causal}} > p_c$ (which appears to be below 0.50 in this system). Control parameter: P_{causal}. Mechanism: reconstructive persistence — zone-specific

signal accumulated in ϕ across many write events, surviving despite (and enabled by) spin volatility.

These two structures are present simultaneously. At $T < T_c$: both attractor persistence (spin order) and reconstructive persistence (phi-order) are active. At $T > T_c$: only reconstructive persistence remains, and it is governed by P_{causal} alone. This is the coexistence of the two persistence strategies at different temperatures, with a clean separation by the Ising phase transition.

5 Universality Class Evidence

The two-substrate comparison (VCML and Ising + VCSM-lite) now supports three criteria for a universality class:

1. **Same control parameter.** Causal purity P_{causal} controls the phi-order in both substrates.
2. **Same order parameter structure.** The phi-order parameter (zone-mean separation normalised by within-zone spread) is structurally identical to VCML's $C_{\text{order}} = S/\sigma_w$.
3. **Orthogonality to standard physical parameter.** In VCML, P_{causal} is orthogonal to the perturbation amplitude. In Ising, P_{causal} is orthogonal to temperature T. In both systems, the purity-controlled transition exists independently of the standard physical control parameter.

What is preserved across substrates is not the dynamics (Metropolis vs. GRU-CML) but the *informational structure*: a slow field accumulates zone-specific signal through causal attribution, and the quality of that attribution — measured by P_{causal} — determines whether the field develops order.

The causal attribution mechanism maps to the universality class, not the carrier. This is the definition of a universality class.

6 The Broader Picture: Why Causal Purity Replaces Temperature

In equilibrium statistical mechanics, temperature controls the ratio of thermal noise to coupling energy, setting whether order can form. In purity-controlled non-equilibrium systems:

Equilibrium stat. mech.	Purity-controlled dynamics
Thermal noise $\sim k_B T$	Causal noise $\sim (1 - P_{\text{causal}})$
Order parameter m	Order parameter S_ϕ
Phase transition at T_c	Phase transition at p_c
Free energy functional	Lyapunov functional C_{order}
Boltzmann factor $e^{-E/k_B T}$	Attribution probability P_{causal}
Equilibrium: thermal ergodicity	Non-equilibrium: causal ergodicity

The "causal temperature" $(1 - P_{\text{causal}})$ plays the same role as thermal temperature but measures *attribution noise*, not energy fluctuations. A system at high causal temperature ($P_{\text{causal}} \approx 0.5$) cannot develop phi-order for the same reason that a ferromagnet above T_c cannot develop spin order: noise overwhelms signal.

This analogy suggests a deep connection: purity-controlled non-equilibrium order may be the information-theoretic counterpart of thermally controlled equilibrium order. The same mathematical structure (order parameter, fluctuation amplitude, critical threshold) appears in both, but the physical quantity controlling the transition differs.

7 Conclusions

1. **Magnetisation m is T-controlled, P_{causal}-independent.** Columns of the (T, P_{causal}) phase table are constant. Standard Ising physics unaffected by the VCSM layer.

2. **Phi-order S_ϕ is P_{causal}-controlled at all temperatures.** All rows increase monotonically with P_{causal}. The two control parameters are orthogonal.

3. **$S_\phi > 0$ at $T > T_c$ (reconstructive persistence survives attractor failure).** At $T = 3.0$ ($m = 0.07$), S_ϕ ranges from 0.51 to 0.91. Zone-specific memory order exists even when the spin substrate is disordered.

4. **Phi-order peaks near and above T_c, not at $T \ll T_c$.** Over-stable attractors (low T, crystallised carriers) starve the reconstructive layer of perturbation signal. Optimal reconstructive persistence requires non-zero carrier volatility. This is the Ising-system analogue of the VCML "death as anti-crystallisation" finding (Papers 77–82–89).

5. **Cross-substrate confirmation of the causal-purity universality class.** Same control parameter (P_{causal}), same order parameter structure (S/σ), same orthogonality to standard physical parameters. The mechanism is substrate-independent: causal attribution quality determines whether reconstructive persistence can operate.

6. **Causal purity as "causal temperature".** $(1 - P_{\text{causal}})$ plays the role of temperature for the purity-controlled transition: it is the noise amplitude, not thermal but informational. The universality class is "purity-controlled non-equilibrium order," distinct from thermally controlled equilibrium order.

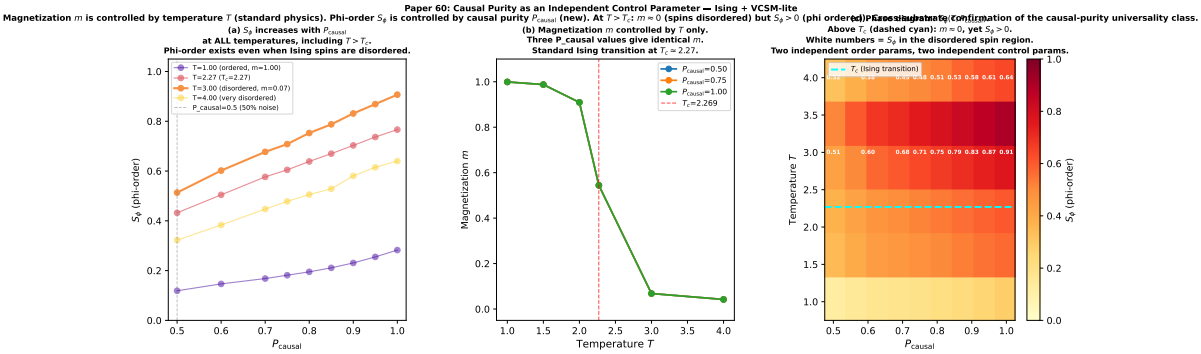

Figure 1: (a) S_ϕ vs P_{causal} at four temperatures. At $T = 3.0$ (Ising disordered, $m = 0.07$), S_ϕ increases from 0.51 to 0.91 with P_{causal} — phi-order exists without spin order. (b) Magnetisation m vs T at three P_{causal} values: identical curves confirm m is P_{causal}-independent. (c) Phase diagram heatmap: $S_\phi(T, P_{\text{causal}})$. Above the cyan dashed line (T_c): spins disordered but $S_\phi > 0$ and growing with P_{causal} (white annotations). Two independent phase structures, two independent control parameters.